Parth Sinha

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Cambridge / London, UK

Personal Statement

Postgraduate researcher specialising in multimodal generative modelling and cross-modal representation learning. My dissertation designs and evaluates a joint Wasserstein Autoencoder with Low-Rank Multimodal Fusion, investigating how separate modalities interact, degrade, and recover within a shared latent space — directly within the scope of multimodal generative models and adaptive multimodal systems. I have implemented custom training pipelines, backpropagation profiling frameworks, and fine-tuned large vision-language architectures on real datasets with rigorous empirical evaluation. I am drawn to Microsoft Research Cambridge's agenda of connecting generative modelling depth with interactive, dynamic environments, and want to contribute to that research at the frontier, through both publications and working systems.

Publications

MSc Dissertation (in progress): **Robustness and Geometry in Low-Rank Multimodal Fusion** — Queen Mary University of London, target submission AAAI 2027.

Education

Queen Mary University of London — London, UK
MSc Computer Science (Artificial Intelligence) — *Sep 2025 – Sep 2026*

- Modules: Machine Learning, Deep Learning, Reinforcement Learning, Natural Language Processing, Information Retrieval, Neural Networks & NLP, Conversational Agents & Dialogue Systems, AI Ethics
- Society: Queen Mary Machine Learning Society

Visvesvaraya Technological University (VTU) — Bengaluru, India
BEng Artificial Intelligence & Machine Learning, CGPA: 8.01/10 (3.3/4.0) — *Dec 2021 – May 2025*

- Modules: Deep Learning, Predictive Analysis, Digital Image Processing, Computer Networks, Blockchain Systems

Research & Projects

MSc Dissertation: Robustness and Geometry in Low-Rank Multimodal Fusion | *PyTorch, IEMOCAP, MOSI*

Sep 2025 – Ongoing

- Designed a joint Wasserstein Autoencoder integrated with Low-Rank Multimodal Fusion (jWAE-LMF) to model how separate data modalities interact within a shared regularised latent space, investigating the geometry of cross-modal representations under varying conditions of signal availability.
- Built an original backpropagation profiling framework to trace tensor fusion mechanics through Hadamard products, providing a mathematical proof of cross-modal gradient coupling via an eight to nine fold increase in gradient sensitivity over LMF baselines under individual modality dropout.
- Applied Integrated Gradients and feature ablation to evaluate forward-pass routing precision, recording a 6.3 percentage point improvement (99.4 per cent versus 93.1 per cent) over additive baselines, then validated on IEMOCAP under a balanced 4× text-to-feature dimensionality regime.
- Designed progressive multi-rate dropout sweeps (zero to ninety per cent) to isolate structural failure modes under severe primary signal loss, generating findings targeted for submission to AAAI 2027.

Custom Vision-Language Assistant (LLaVA) | *PyTorch, LongCLIP, Qwen3, LoRA/PEFT*

Mar 2026 – May 2026

- Engineered a custom multimodal generative architecture aligning a LongCLIP-ViT-B-32 vision encoder with a Qwen3-0.6B language backbone via a purpose-built multimodal projection layer, mapping visual tokens directly into the language model's embedding space.
- Designed a vector semantic alignment pipeline to reconstruct unindexed image-to-MCQ mappings across 11,600 images and 65,000 questions, achieving over 99 per cent alignment accuracy through controlled fine-tuning rather than brute-force data expansion.
- Applied LoRA for parameter-efficient fine-tuning (PEFT) alongside task-oriented prompt engineering and image augmentation, operating the full pipeline within hardware-constrained compute while preserving generation quality.

Hybrid Document Retrieval System and DPR Search Engine | *Python, FAISS, ColBERT, BM25, DistilBERT*

Mar 2026

- Designed a multi-stage retrieval architecture combining sparse (BM25 with RM3) and dense (TCT-ColBERT) retrieval with FAISS indexing and Reciprocal Rank Fusion, formally evaluated against the BEIR benchmark.
- Built a Dense Passage Retrieval engine using dual-encoder DistilBERT with in-batch negative sampling, supporting offline indexing and online query-time retrieval, engineered as a clean modular codebase rather than a notebook prototype.

Credit Card Fraud Detection using GANs (Published) | *International Conference on Data Science and Mining*

Jan 2024 – May 2025

- Designed Generative Adversarial Networks for real-time fraud detection on high-throughput transactional pipelines, diagnosed mode collapse under severe class imbalance, and redesigned as a Wasserstein GAN to stabilise training and normalise synthetic sample generation.
- Achieved 95 per cent testing accuracy on downstream classification, validating the generative model's fidelity on unbalanced real-world data. Published at the International Conference on Data Science and Mining.

Technical Skills

Research Frameworks: PyTorch, TensorFlow, Hugging Face Transformers, JAX (familiar)
Multimodal & Generative AI: LLaVA, LoRA/PEFT, GANs (WGAN), latent space regularisation, cross-modal fusion architectures
Retrieval & Agentic Systems: LangChain, FAISS, BM25, ColBERT, RAG architectures, multi-agent orchestration
Training Infrastructure: Distributed and parallel computing (MPI), multi-GPU training, CI/CD (GitHub Actions), Docker, AWS
Languages: Python (advanced), Java, Rust, SQL, C
Evaluation & Analysis: Integrated Gradients, backpropagation profiling, feature ablation, BEIR benchmarking, TensorBoard

Competitions & Leadership

Market Making Hackathon, 6th Place (Highest Solo Entrant) | *QMUL Machine Learning Society*

Mar 2026

- Built a multi-agent autonomous trading system with a hierarchical routing mechanism between a regime switching model, a Q-learning agent, and an XGBoost ensemble, finishing as the sole solo entrant in the top ten against 95 or more teams.

St. Valentine's Hackathon, 1st Place | *QMUL Machine Learning Society and QMUL Fintech Society*

Feb 2026

- Built an XGBoost and LightGBM ensemble classifier with advanced feature engineering, securing first place out of 24 or more competing teams under strict time constraints.

Founder & Former Head, Saragam Music Club by AIML (VTU) Jun 2023 – May 2025
- Founded and led a campus music organisation, organising large-scale events and weekly sessions to build community and collaborative culture.

Professional Experience

Part-Time STEM Tutor Apr 2026 – Present
Tutopiya.com *London, UK*
- Deliver 1-to-1 tuition in Mathematics, Computer Science, and Physics, distilling abstract technical concepts into structured explanations for varied audiences — directly relevant to Distilling research insights into presentations and written communications.

Emerging Technologies (AI & Cloud) Intern Oct 2023 – Nov 2023
Edunet Foundation / IBM *Bengaluru, India*
- Deployed a Watsonx Assistant on IBM Cloud infrastructure, designing conversational logic and NLU components to interpret complex, multi-part specifications accurately.

References

Available upon request.